

POLICY NUMBER: CP02.111

CATEGORY: Patient Care

REVIEW RESPONSIBILITY/DOCUMENT OWNER: Clinical Nurse Specialist

TITLE: In-hospital Extracorporeal Cardiopulmonary Resuscitation (ECPR) in UF Health Heart & Vascular and Neuromedicine Hospital (HVN)

POLICY: Extracorporeal Cardiopulmonary Resuscitation (ECPR) has the potential to improve prognosis for a subset of in-hospital cardiac arrest patients during CPR and/or as part of CPR stabilization, including interventions to reverse the etiology of the arrest.

ECPR requires certain resources (ECMO Capacity and Medical Staff) in immediate proximity (time and place) and seamless alignment to include ECMO consideration routinely into clinical emergency resuscitation. The resources currently available in and to the HVN align to form a foundation for an inpatient ECPR pilot program at UF Health Shands. From this foundation, there is potential to implement ECPR in other Hospitals at UF Health Shands.

PURPOSE: This document outlines details of the in-patient Extracorporeal CPR (ECPR) program for a subset of patients in refractory in-hospital cardiac arrest. It describes notification and alert criteria and roles & responsibility of those responding.

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DEFINITIONS:

- A. Extracorporeal CPR (ECPR) – the implantation of veno-arterial extracorporeal membrane oxygenation (VA-ECMO) in a patient who experienced a sudden and unexpected pulseless condition attributable to cessation of cardiac mechanical activity. ECPR is recognized in Extracorporeal Life Support Organization (ELSO) and American Heart Association (AHA) guidelines as a technique that can be considered in select patients in cardiac arrest.
- B. ERS attending – Emergency Resuscitation Service attending.
- C. Attending Physician: The physician primarily responsible for the patient's care.

CORE PROCEDURE:

I. Determination of ECPR Candidate

- A. Primary service or code leader will identify patients who meet inclusion criteria (see [Appendix A](#) for table of inclusion/exclusion criteria).
 - 1. Special consideration should be made after 2 rounds of Advanced Cardiac Life Support (ACLS).
 - 2. A code leader may be any CCM attending, fellow, or critical care APP.
 - 3. Upon identifying a potential candidate, the primary service or code leader will call the ERS attending on-call (352) 283-4198.
 - 4. If a patient does not meet criteria, the ERS Attending or CT surgery fellow/attending may still be called at the discretion of the primary service for consideration of ECPR.
 - 5. ERS attending may make an independent decision to cannulate patients. Primary service does not need to sign off on the ECPR.
- B. The following individuals can place the call to the ERS attending: Any critical care APP, any physician, any ECMO specialist, and any RN. Attending to attending communication is preferred.
 - 1. After discussing the patient, the ERS attending will decide on initiating the in-patient ECPR protocol.

II. Response for ECPR Patient

- A. Communication across stakeholders (See [Appendix B](#): Important Contacts)
 - 1. The ERS attending will call the ECMO Primer to bring supplies to bedside.
 - 2. The ERS attending will call the Charge RN on either 77 or 87 to facilitate communication to the nursing coordinators to obtain an ICU bed for the ECPR patient.
 - 3. The goal is to mobilize the resources to assign an ICU bed and to prepare for transport from the coding unit after patient is stabilized.
 - 4. ECMO initiation will then proceed in the coding unit, with the intent that patients will

proceed rapidly to the appropriate ICU once cannulated.

5. If the patient is deemed to be a candidate for ECPR cannulation, both primary team and cannulating physician will agree on cannula configuration and on which team will remain/take over as primary team prior to patient transfer to ICU.
6. Should it be deemed clinically appropriate, the ERS Attending will discuss the patient with the on call interventional cardiology attending for the consideration of coronary angiography after ECMO is initiated.

B. Cannulation of ECPR Candidate

1. The CCM team should be actively working on placing bilateral arterial and venous femoral lines to assist with access while cannulator arrives.
2. Supplies (See [Appendix C](#): Cannulation Supplies)

C. Response to Medical Surgical Units (see [Appendix D](#): HVN Med Surg ECMO Response Plan for specifics by location)

1. Code Blue Response Team (potentially including roles described in other sections of this policy) is likely already caring for the ECPR candidate. See Core Policy CP02.007 Clinical Emergency Response Plan.
2. Primary ICU CN responds with access kit and procedure cart*; backup ICU CN will back up Primary ICU CN as necessary.
3. Primary ICU attending physician communicates with patient's primary attending physician regarding the code and next steps.
4. If ECMO is needed, Primary ICU CN to call ECMO primer.
5. ECMO primer to bring ECMO cart and circuit/pump to bedside.
6. Primary ICU attending contacts cannulating attending if different from primary attending.
7. Stat nurses communicate bed placement needs to nursing coordinators.
8. **If ECMO is initiated on MS floor- A provider designated by the ERS attending and the ICU CN will remain with patient until ICU transfer is complete.**

D. ICU Transfer

1. Patient will be transferred to an ICU bed as soon as possible.
2. A bed may be held open on a designated ICU for ECPR events.
 - a. In the event a patient takes the last open ICU bed, a patient will immediately be identified and relocated out of the unit to restore an open bed designated for ECPR.
 - b. The relocation could include a patient downgrade, movement to an off-service ICU, or boarding in PACU.

III. Initiation of ECMO

- A. ECMO Capacity: The in-house ECMO primer will oversee the fluid nature of the organizations ECMO census and will be in ongoing communication with the ERS attending and process stakeholders (See [Appendix E](#): ECMO Capacity).
- B. If ECMO is initiated on MS floor- A provider designated by the ERS attending and ICU CN will remain with patient until ICU transfer is complete.

IV. Quality control

- A. Case review will be conducted in the ECMO Quality meetings and reported through QPSR and CERC annually.

ASSOCIATED POLICIES:

CP02.007 Clinical Emergency Response Plan

Applicable ECMO Policies

KEY WORDS: CPR, ECPR, ECMO, Cannulation, ERS

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Final Audit Report

2024-11-05

Created:	2024-11-04
By:	Nisha Patel (pateln@shands.ufl.edu)
Status:	Signed
Transaction ID:	CBJCHBCAABAA_VJVq8UKcl4LMq2jPhAaOTd3rfJLUHpX

"CP02.111 In Hospital CPR" History

-  Document created by Nisha Patel (pateln@shands.ufl.edu)
2024-11-04 - 9:20:55 PM GMT
-  Document emailed to Dr. Tim Morey (tmorey@anest.ufl.edu) for signature
2024-11-04 - 9:20:59 PM GMT
-  Document emailed to Dr. Steve Motew (steve.motew@ufhealth.org) for signature
2024-11-04 - 9:20:59 PM GMT
-  Email viewed by Dr. Steve Motew (steve.motew@ufhealth.org)
2024-11-04 - 10:10:06 PM GMT
-  Document e-signed by Dr. Steve Motew (steve.motew@ufhealth.org)
Signature Date: 2024-11-04 - 10:10:16 PM GMT - Time Source: server
-  Email viewed by Dr. Tim Morey (tmorey@anest.ufl.edu)
2024-11-05 - 0:57:19 AM GMT
-  Document e-signed by Dr. Tim Morey (tmorey@anest.ufl.edu)
Signature Date: 2024-11-05 - 0:57:40 AM GMT - Time Source: server
-  Agreement completed.
2024-11-05 - 0:57:40 AM GMT